

Vineet Agarwal

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Summary | Software engineering grad student with 2+ years production experience, focused on multi-agent AI systems and LLM inference optimization, from agent orchestration (LangChain, LangGraph, Bedrock) down to speculative decoding and multi framework inference benchmarking on GPU workloads.

Skills

LLM Inference & Optimization vLLM, SGLang, TensorRT-LLM, llama.cpp, speculative decoding, KV cache

Applied AI LangChain, LangGraph, AutoGen, RAG pipelines, OpenAI APIs, Amazon Bedrock, Hugging Face

Databases PostgreSQL, MongoDB, Redis, FAISS, Pinecone

Cloud & DevOps AWS (Lambda, EKS), Docker, Kubernetes, CloudFormation, GitHub Actions CI/CD

Languages Python, JavaScript, TypeScript, C++, SQL, Go

Frontend React, Next.js, React Native, Redux Toolkit

Backend FastAPI, Node.js, Express, Django, REST APIs, Web-Socket, WebRTC

Core Engineering Data Structures & Algorithms (C++), OOP, System Design, Microservices, Agile/Scrum

Experience

ServBeyond Solutions

Jun 2026 – Present

Enterprise AI & Platform Intern

- Built and shipped a RAG assistant over internal documentation using LangChain and OpenAI APIs, cutting 25 hours of manual lookup per week and reaching 95% answer accuracy across 100 users.
- Designed agentic GenAI workflows across Salesforce, ServiceNow, and AWS using Amazon Bedrock for agent orchestration and REST plus SOAP integration into a normalized data layer.

Runara.ai

Apr 2026 – May 2026

ML & Inference Engineer Intern

- Implemented speculative decoding from scratch across two GPUs in PyTorch and HuggingFace Transformers, applying the paper's probabilistic rejection sampling criterion rather than naive argmax matching.
- Diagnosed a gap between 92% isolated draft agreement and 29% in-loop acceptance, tracing it to KV cache and verification input handling and showing argmax draft selection beats temperature sampling.
- Benchmarked Qwen3-Coder-480B across vLLM, SGLang, and TensorRT-LLM on RTX Pro 6000 Blackwell GPUs from 1K to 128K context, profiling TTFT, throughput, KV cache growth, and multi user scalability.

Svipes

Jul 2024 – Dec 2024

Software Engineer

- Shipped 50+ React Native screens with Redux Toolkit and cut video playback stutter 70% via async loading and client side caching.

Xelpmoc Design and Tech Limited

Nov 2022 – Apr 2024

Software Engineer

- Reduced API response time from 75s to under 10s and increased throughput 4x by refactoring SQL joins, indexing high-traffic tables, and adding a Redis caching layer.
- Architected RESTful backend services for a tourism platform serving 10K+ daily bookings; cut production defects 40% and accelerated feature delivery 25%.

IIIT Hyderabad

Jun 2022 – Aug 2022

Software Intern

- Automated multilingual data extraction across 5+ regional languages with Python and Selenium; lifted NLP classification accuracy from 78% to 93%.

Education

University of Maryland, College Park

Jan 2025 – Dec 2026 (Expected)

Master of Engineering, Computer Software Engineering, Minor in Cloud Engineering

GPA: 3.7

Projects

RepoResearchAI: Multi-Agent Code Analysis Platform (github.com/vineetagarwal54/RepoResearchAI)

- Built a multi-agent AI system with AutoGen, LangChain, and FAISS vector search; specialized agents (SDE, PM, QA) generate architecture insights, API docs, and schema analysis using GPT-4 with RAG over codebase embeddings.

Care Bridge AI (github.com/heena5498/carebridge-ai)

- Built a healthcare assistant at Bitcamp using FastAPI, PostgreSQL, React, and the Gemini API, delivering conversational patient support with low-latency responses.